
The Summary Never Reads the Mask: Localisation-Grounded Report Faithfulness in a Deployed Chest Radiograph System

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Abstract

1 Deployed chest radiograph systems increasingly show a diagnostic prediction,
2 a localisation overlay, and a short natural-language summary together, and the
3 overlay is read as evidence that the summary is grounded in the image. We audit
4 one such deployed four-class system (Normal, pneumonia, tuberculosis, COVID-
5 19; DenseNet-121 classifier, Grad-CAM localisation compressed into a U-Net for
6 one-forward-pass serving). Its summary generator turns out to be a four-entry
7 static template keyed only on the predicted class: no property of the mask, not the
8 side, the region, the extent, nor even whether a region exists, reaches the printed
9 text. Over the system’s 245 already-scored test overlays we then show, with no
10 retraining, that gating the summary on measured mask properties changes what
11 can honestly be said; that on most films the honest output is to say nothing about
12 location, because the overlay sits outside the lung fields; and that permuting each
13 film’s mask onto a different film drops claim accuracy to near chance (12% on
14 average over 1,000 random permutations). That last result is the failure mode
15 the current template’s near-silence on location happens to sidestep and the one a
16 naive descriptor-driven generator would walk straight into. We report this on a
17 deployed system and give a small evaluation protocol, crossing heatmap source
18 with generator type, for the general question of when a generated radiology report
19 is reading the image it describes.

20 1 Introduction

21 Chest radiograph systems built for low-resource triage bundle three outputs behind one click: a
22 diagnostic label, a heatmap that claims to show where the evidence lies, and a short clinical-language
23 summary. Each is usually evaluated on its own, if at all. Classification accuracy is scored against
24 a held-out set, localisation quality against some overlap reference, and the sentence is treated as
25 presentation. That split hides a specific, checkable failure: a system can be accurate, can show a
26 heatmap that looks like an explanation, and can still print a summary whose spatial claims, when it
27 makes any, land on the mediastinum or the cardiac silhouette instead of the lung.

28 We audit a system with exactly that shape. LungLens is a four-class chest X-ray classifier on a
29 DenseNet-121 backbone [Huang et al., 2017], with Grad-CAM [Selvaraju et al., 2017] localisation
30 compressed into a U-Net [Ronneberger et al., 2015] so an overlay costs one forward pass at serving
31 time, not a forward-plus-backward pass. Every prediction carries a summary. Reading the source
32 settles what writes it: a four-entry lookup keyed on the predicted class, joined to one of five fixed
33 sentences chosen by whether the prediction is confident, uncertain, or Normal. No coordinate, area,
34 side, or component count from the mask is ever an input. Nothing else feeds it, so the question

35 of whether the localisation subsystem contributes anything to the report has a direct answer, and
36 Section 4 is built to give it.

37 We turn that question into a measurement. Reusing the region-level statistics the system’s own failure
38 analysis already computes over 245 scored test films (side, in-lung fraction, area fraction, plus a
39 component count we add), we build a small descriptor-to-prose pipeline and compare it with the
40 deployed template under four heatmap conditions (no mask, the served Grad-CAM map, the served
41 U-Net map, and a mask lifted from a different film) crossed with two generators: the static template
42 and one gated on the descriptors. The in-lung gate suppresses a spatial claim on 62% of films, because
43 the served overlay’s centre of mass falls outside the lung fields on a majority of them. Permuting the
44 mask onto a different film leaves the claim rate unchanged but drops the claim’s accuracy against the
45 film’s own ground truth to 12% on average over 1,000 permutations, the sanity check the saliency
46 literature uses to tell a method that reads the input from one that does not [Adebayo et al., 2018].

47 **Contributions.** (1) An audit, from source, of what a deployed chest-radiograph summary generator
48 conditions on, and a direct answer to whether its localisation subsystem reaches the text. (2)
49 A retraining-free descriptor pipeline over the system’s own overlays with an in-lung gate, and a
50 measurement of how often that gate fires on real served overlays. (3) A degenerate-mask sanity check
51 in the tradition of Adebayo et al. [2018], applied to the generated report rather than the saliency map
52 alone. (4) A small evaluation protocol crossing heatmap source with generator type, cells we could
53 fill without new training or clinician time filled and the rest left open.

54 2 Related work

55 **Classification and its explanations.** CNN interpretation of chest radiographs matured with CheXNet
56 [Rajpurkar et al., 2017], a 121-layer DenseNet at radiologist level for pneumonia on ChestX-ray14
57 [Wang et al., 2017]; DenseNet-121 [Huang et al., 2017] stays competitive with residual [He et al.,
58 2016] and compound-scaled [Tan and Le, 2019] backbones at chest-radiograph data scales, which is
59 why it is a fixed component here, not a contribution. Four-class Normal/pneumonia/TB/COVID work
60 exists [Chowdhury et al., 2020, Kc et al., 2021], as do multi-task designs sharing a representation
61 across classification and segmentation [Gündel et al., 2019]. Grad-CAM [Selvaraju et al., 2017] and
62 its variants [Chattopadhyay et al., 2018] are the default explanation for such models [Singh et al., 2020],
63 and a body of work asks whether the maps are faithful to the model or only plausible: Adebayo et al.
64 [2018] show several saliency methods survive randomising the model’s weights, and Kindermans
65 et al. [2019] extend the concern. Our degenerate-mask control carries that test from the map to the
66 report built on it: if an unrelated map leaves the text’s claims as accurate, the map is not doing the
67 work the text implies.

68 **Generated reports.** A separate line generates full narrative reports from radiographs [Jing et al.,
69 2018, Chen et al., 2020], usually scored with text-overlap metrics that reward fluency without
70 checking image support. Factual-consistency metrics over clinical entities [Miura et al., 2021],
71 region-grounded generation that ties each sentence to an image region [Tanida et al., 2023], and
72 biomedical vision-language pretraining [Bannur et al., 2023] all respond to that gap. LungLens
73 predates this: its summary is closer to a caption, and we ask the same faithfulness question at the
74 point a much simpler system first gets the chance to fail it. Earlier hand-crafted CXR features [Jaeger
75 et al., 2014, Cortes and Vapnik, 1995, Breiman, 2001] and the COVID-era CNN surveys [Sethy et al.,
76 2020, Wang et al., 2020, Apostolopoulos and Mpesiana, 2020, Ozturk et al., 2020, Minaee et al.,
77 2020] are background for the backbone and dataset choices below.

78 3 System

79 **Data.** The classifier trains on eight public chest-radiograph sources, deduplicated by content hash to
80 37,553 images and sampled to 32,000, arranged so each of the four classes draws from at least two
81 acquisition pipelines, a deliberate hedge against the source-fingerprinting failure of Zech et al. [2018];
82 Section 6 reports the residual. Images are border-cropped 8% at train and inference time to drop
83 burned-in corner text, resized to 224×224 , and ImageNet-normalised. The split is patient-grouped
84 and reproducible: because every directory listing is sorted before use, re-running split construction
85 from the raw files reproduces the test accuracy to four significant figures on the same 4,759 held-out

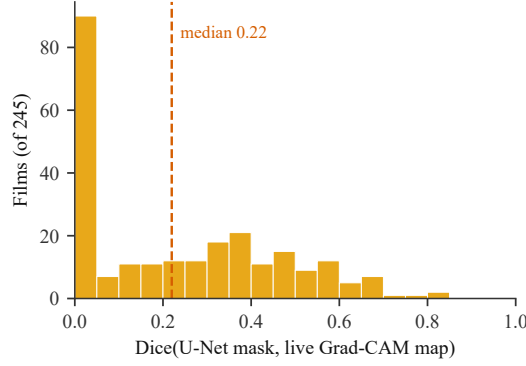


Figure 1: Dice between the served U-Net mask and a freshly computed live Grad-CAM map, same film and class, over all 245 served films. The distribution is heavy at zero, not merely low on average: a generator that reads a location from one of these two sources cannot assume the other agrees with it.

86 images. That guarantee is not free. An earlier, unsorted version of the same code moved the reported
87 test accuracy by nearly two points purely as a function of filesystem directory-traversal order.

88 **Classifier.** A DenseNet-121 [Huang et al., 2017], ImageNet-initialised, fine-tuned under class-
89 weighted cross-entropy (inverse-frequency weights, mean 1.0) for up to 40 epochs with early stopping
90 on validation macro-F1, Adam at 10^{-4} , batch 32. It is a fixed upstream component; its full evaluation
91 is in Appendix A.

92 3.1 Localisation: explanation amortisation, not knowledge distillation

93 A second network, a U-Net-style encoder-decoder (MultiTaskUNet: four DoubleConv blocks,
94 $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ channels, skip connections, 7,705,224 parameters, counted from the
95 served checkpoint), learns to reproduce the classifier’s Grad-CAM map in one forward pass, avoiding
96 the backward pass Grad-CAM needs. The frozen DenseNet (no gradient enters it here) produces
97 a Grad-CAM map at `denseblock4.denselayer16.conv2` for each image’s *ground-truth* label,
98 ReLU’d, resized to 224×224 , and divided by its own maximum; targets are held in memory for the
99 run and never cached to disk. The U-Net is trained against that target with an unweighted Dice+BCE
100 loss (Adam, 3×10^{-4} , batch 8) over 3,000 images for 20 epochs; the served channel is the class
101 being visualised.

102 We call this *explanation amortisation*, not knowledge distillation. Nothing is shared between the two
103 networks: no layers, no logit or feature matching, no gradient path from student to teacher. The only
104 channel is one 224×224 per-class scalar field, computed once and discarded. Very little survives a
105 channel that narrow, and the cost is measurable: on the 245 served films, the median Dice between
106 the U-Net’s mask and a fresh live Grad-CAM map for the same film and class is 0.22, with 27%
107 showing no overlap and 15% above 0.5 (Figure 1). What the compression buys is a real serving-cost
108 saving (Section 5); stating that saving together with this cost is the honest description.

109 3.2 Report generation

110 The version under review disposed of this subsystem in eleven words. Section 5’s central finding
111 needs its exact behaviour, so we specify it here.

112 **What it is.** A static Python dictionary keyed on the predicted class (Normal, Pneumonia, Tuberculosis,
113 Covid-19), joined to one of five fixed sentences selected by three booleans: is the prediction below
114 the confidence threshold, does the visualised class equal a confident non-Normal prediction, and was
115 a mask computed at all. No model, prompt, decoding parameters, or learned weights; the full listing
116 is Appendix B.

117 **Inputs.** The argmax of the softmax output and its probability; a flag for whether the visualised class
118 is the top prediction; a flag for whether that top prediction is Normal; and, for the trailing provenance
119 line only, which subsystem produced the mask. No coordinate, area, side, zone, or component count
120 from the mask is ever read.

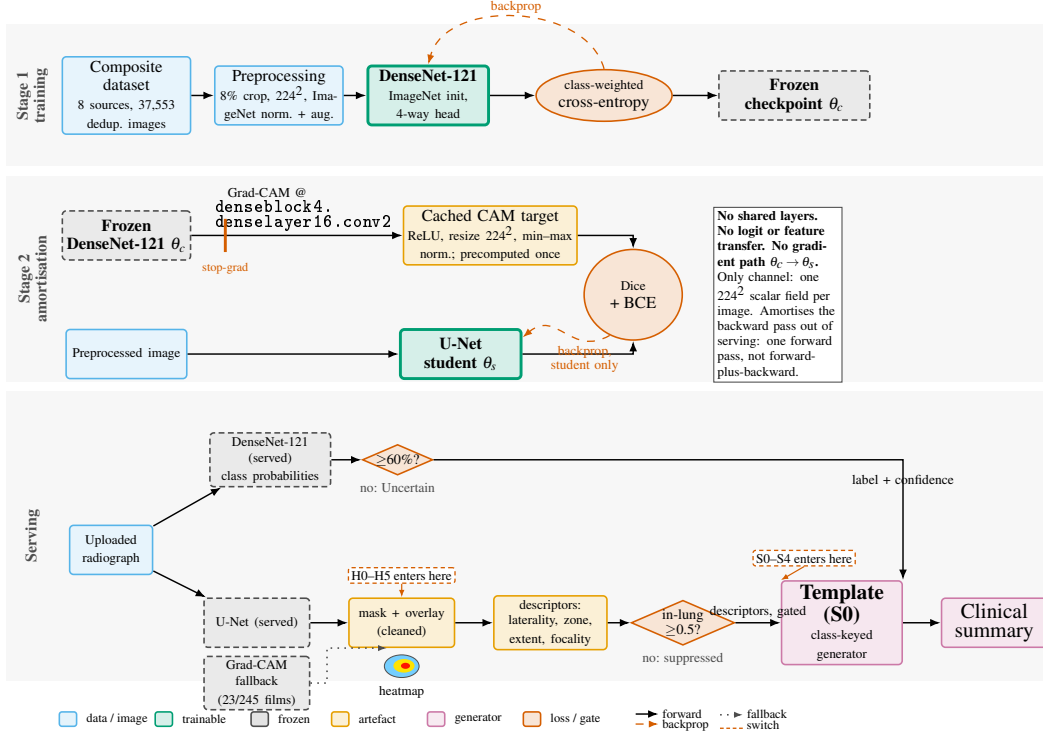


Figure 2: System overview. Stage 1 trains the served DenseNet-121 classifier. Stage 2 supervises a U-Net student on Grad-CAM targets cached from the frozen classifier (stop-gradient marked): no layers, logits, or features are shared, so the only channel is one 224×224 scalar field per image, amortising the backward pass out of serving. At inference the classifier supplies label and confidence; the localiser supplies gated region descriptors; both arrive at the report generator, a class-keyed template (S0), which alone determines the printed summary. Tags mark where H0–H5 and S0–S4 enter.

121 **Confidence gate.** Below 60% top-class probability the report reads “Uncertain” with a fixed
122 cautionary sentence; the only other non-standard path is a caught exception on malformed input.

123 **Failure this produces.** The “confident non-Normal” branch does not check whether a region survived
124 the display threshold, so the text can assert “the outline marks the most influential area” on a film
125 where nothing was outlined. This happens on 2 of the 245 scored films, one a confidently wrong
126 tuberculosis call (95.7%) on a radiograph whose true label is Normal, with no thresholded region
127 anywhere in the image (Appendix C).

128 **A location fixed by class, not by image.** One branch does name a location: every confident Covid-19
129 prediction prints “possible bilateral ground-glass opacities”, whatever the overlay shows. On the 245
130 scored films this sentence appears 10 times; the served region actually spans both lung fields on 6 of
131 them. The word is a class prior, not a reading of the image, and it is the only spatial claim S0 ever
132 makes.

133 4 Experimental design

134 Classification accuracy and localisation quality are already measured (Section 3, Appendix A). The
135 new experiment asks one thing: does the summary change, and change correctly, as a function of the
136 heatmap it is given. We cross a heatmap condition with a generator condition over the 245 test films
137 already scored for the failure analysis (all 102 pneumonia-predicted-Normal errors, all 92 Shenzhen
138 and 12 Montgomery test films, and 40 stratified correct films).

139 **Heatmap conditions.** H0, no mask (class and confidence only). H1, the live Grad-CAM map. H3,
140 the served U-Net map. H5, a degenerate control: the mask of a uniformly random *different* film under

Table 1: The evaluation grid. Filled cells report a measured number (Section 5); open cells are proposed and not run. S0’s text is identical under every heatmap, so its row collapses by construction, which is the headline; its only spatial word is the class-fixed “bilateral” in the Covid-19 sentence. S1’s heatmap cells are reported jointly as H_actual; a per-film H1-vs-H3 split is left for a revision.

Generator (rows) / Heatmap (cols)	H0 (none)	H1 (Grad-CAM)	H3 (U-Net)	H5 (permuted)
S0 (deployed template)	measured	measured	measured	n/a (static)
S1 (descriptor template)	measured	measured	measured	measured
S2 (text LLM)	open	open	open	open
H2 (Grad-CAM++), H4 (second backbone), S3/S4 (VLM): open, see Section 6				

a fixed derangement (seed 42), so each film keeps its own class and confidence but gets someone else’s spatial evidence.

Generator conditions. S0, the deployed generator (Section 3.2), rebuilt from source. S1, a template stating side, vertical third, extent, and focality from the region descriptors, gated to make no claim when the thresholded region’s in-lung fraction is below 0.5 or when no region survives the display threshold. S2, a text-only LLM given class, confidence, and descriptors, is out of scope here (Section 6).

Region descriptors. For each film the served overlay’s thresholded region (the app’s own thresholds and cleaning: 0.50 for a U-Net mask, 0.45 for Grad-CAM, 5×5 elliptical opening, components under 0.3% of the frame dropped) is reduced to laterality (left/right/outside both, from the centroid against a separate lung-field segmenter), vertical zone (upper/mid/lower third, from the centroid; new here), extent (focal <5%, patchy 5–25%, extensive >25%, from area fraction), focality (single vs. multifocal, from component count; new here), and in-lung fraction (the gate). Laterality, extent, and in-lung fraction were already in the system’s analysis code; the two new descriptors were checked against it by matching recomputed area fractions to a median absolute difference of 0.0001 over the 204 films with a region.

Cells run (no GPU, no training): S0 as baseline; S1 crossed with H0, H_actual (whichever of H1/H3 the system served for that film), and H5. A per-film H1-versus-H3 pairing, computing both maps fresh for every film, needs the 245 radiographs staged for a forward-plus-backward pass each and is left for a revision.

Metrics. Spatial-claim rate (how often a claim is made). Grounding precision (does a claim match the film’s descriptors). Phantom-claim rate (does the generator assert evidence that did not survive the threshold). Anatomical plausibility (does a claimed location fall in the lung, true by construction under the gate). H5 grounding accuracy (does a claim from another film’s descriptors match this film’s truth). Latency, reported in prose.

5 Results

Classifier, in one paragraph. The DenseNet-121 reaches 96.18% accuracy and 95.77% macro-F1 on the patient-grouped 4,759-image test set, with per-class recall from 92.2% (pneumonia) to 98.5% (Normal). The dominant error, 102 pneumonia films predicted Normal (56% of all 182 errors), sits in a single source folder whose label, *lung opacity*, is a broader radiological finding than pneumonia, and several of those films have clear lung fields on inspection: label noise as much as model failure. Full detail is in Appendix A.

5.1 The summary does not move with the mask, because it never reads it

Under every heatmap condition (H0, H1, H3, H5), S0 prints byte-identical text for a given film, because none of its five branches take mask geometry. The explainability apparatus and the report apparatus are disconnected. In Table 1 this is the collapsed S0 row: a property of the generator, not of our harness.

S0 (deployed). Prediction: Covid-19 (99.1%). Findings consistent with Covid-19. Warmer areas indicate possible bilateral ground-glass opacities. Heatmap: region driving the Covid-19 prediction. The outline marks the most influential area. *Overlay source: disease-segmentation U-Net.*

S1 (descriptor, H_{actual}). Prediction: Covid-19 (99.1%). Findings consistent with Covid-19. Warmer areas indicate possible bilateral ground-glass opacities. Heatmap evidence: left lung, mid zone, patchy, multifocal. *Overlay source: disease-segmentation U-Net.*

(Reconstructed from the template in Appendix B and the film’s measured descriptors; *figures/paired_summary.pdf* overrides.)

Figure 3: The same radiograph, predicted Covid-19 at 99.1% confidence, under both generators. S0 is identical whether or not a mask is available; S1 adds a spatial claim built from the film’s own descriptors. Both are language over measured pixels, not a clinical finding.

5.2 Gating on measured evidence changes what can honestly be said

S1, gated on in-lung fraction, states a spatial claim on 92 of 245 films (37.6%). On the other 153, either no region survives the display threshold (41 films, 16.7%) or the region’s centre of mass is outside the lung fields. The gate rate follows directly from an already-published measurement: 55% of the served overlays put the majority of their region outside the lung (Appendix D). Figure 3 shows a pair on one film: S0’s class-fixed text above, S1’s grounded “left lung, mid zone, patchy, multifocal” below, read from the film’s own descriptors. Without localisation the generator can only restate the class; with it, it can also say where, and the accuracy of that “where” is bounded by Section 3.1’s Dice.

5.3 The degenerate-mask control

Permuting each film’s mask onto a different film under a fixed derangement leaves S1’s claim rate unchanged (92/245, since a permutation preserves the marginal of which films pass the gate), but the claim’s accuracy against the film’s true side, zone, and extent falls to a mean of 12.0% over 1,000 random derangements (95% of derangements between 6.5% and 18.5%), near the chance level the skewed zone and side marginals imply. A single derangement is too noisy to quote alone: the one in our first draft gave 14/92 (15.2%), inside that range. This is Adebayo et al. [2018]’s check moved from the saliency map to the report a naive descriptor-driven generator would write: if the stated location is no more accurate with the right mask than the wrong one, the localisation subsystem is not grounding the sentence. The same bound acts on any real generator drawing from H1 or H3: the two maps disagree at median Dice 0.22 (Section 3.1), so a generator reading one of them inherits that ceiling. A localisation-grounded report is exactly as trustworthy as the localisation under it, and no more.

Latency. On a CPU laptop (i7-13620H, 10 threads), the served pipeline averages 293 ms (sd 81) when the distilled U-Net supplies the mask, the common case at 91% of test films, and 560 ms (sd 22) on the live Grad-CAM fallback (forward-plus-backward, 9%), a test-set-weighted 318 ms. The roughly 2× gap is what the U-Net buys on hardware with no GPU. Report generation is a dictionary lookup and adds nothing measurable; a hosted-API generator would need its network latency reported separately.

6 Proposed protocol and limitations

What is proposed, not measured. Table 1’s open cells are not implied results. H2 (Grad-CAM++) and S3/S4 (a vision-language model given the radiograph, or the radiograph plus the overlay) are inference- or API-cost work scoped out by design. H4, Grad-CAM from a second, independently trained backbone, is the most informative open cell, since it separates “the summary tracks the pathology” from “the summary tracks this model,” and it needs a second classifier trained end to end. No clinician was available; the automated metrics are faithfulness proxies, not substitutes. An LLM-as-judge score, if used later, must report its agreement with human labels on a subsample or not be used.

Dataset-source bias. Despite multi-source classes and identical 8% cropping, per-source test accuracy runs from 83.3% (Montgomery, 12 images) to 99.5% (TBX11K). The two weakest sources fail oppositely in the localisation data: Shenzhen’s overlays are diffuse (median 41% of the frame on TB-positive films), Montgomery’s are mostly absent (a quarter of its films produce no region). Source and class stay partly confounded in an eight-source composite, so 96.18% is an optimistic upper bound, not a clean prospective number.

Label noise. 82 of the 102 pneumonia-to-Normal errors come from a folder whose label, *lung opacity*, is a mapping decision rather than a diagnosis, and several inspected films have clear lung fields. Some of this error mode is model failure and some is the model correctly reading a film that should not have been labelled positive; the two are not separable after the fact without the original read.

Reproducibility. An earlier, unsorted directory-traversal order moved the split from 22,419/4,785/4,796 to 22,420/4,759/4,821 images and a re-scored checkpoint from 96.62% to 98.11%, nearly two points from enumeration order alone on an otherwise identical pipeline and seed. Sorting every listing before use is now load-bearing for every number here being reproducible.

No clinical validation. LungLens is a research and educational system, not evaluated prospectively against radiologist reads. Its summary, static template or descriptor-gated, is language over measured pixels, not a clinical finding, and the deployed interface should say so, not only this paper.

7 Broader impact

A confident label, an apparently explanatory heatmap, and a fluent summary shown together are more persuasive than their parts justify, especially to a non-specialist in the low-resource setting this system targets. Our central finding, that a plausible summary can be entirely disconnected from the localisation evidence beside it, is a caution against that persuasiveness, not an endorsement of shipping the descriptor-gated version as a clinical tool. Any future report layer here should carry the non-diagnostic disclaimer this paper argues is missing, and be checked for faithfulness before fluency.

8 Conclusion

LungLens pairs an accurate classifier, a fast localisation shortcut, and a clinical-language summary. Auditing the third from source shows it never reads the first two: the summary depends only on the predicted class and its confidence. A small retraining-free gate on the system’s own region statistics shows what changes when it does read them, and most of the time the honest output is to say nothing about location, because the evidence sits outside the lung. The degenerate-mask control makes the point: without a faithfulness check, a more sophisticated report generator would not be safer, only more convincing.

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328 Recognition (CVPR)*, pages 2097–2106, 2017.
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330 Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A
331 cross-sectional study. *PLoS Medicine*, 15(11):e1002683, 2018.

332 A Classification detail

333 The classifier is a fixed upstream component (Section 3); this is its full evaluation.

Table 2: Per-class performance of the served DenseNet-121 on the held-out test set ($n = 4,759$; accuracy 96.18%, macro-F1 95.77%). Percentages except support. Derived from the confusion matrix below.

Class	Prec.	Recall	F1	Spec.	Supp.
Normal	95.30	98.50	96.90	94.60	2,502
Pneumonia	98.30	92.20	95.10	99.30	1,478
Tuberculosis	94.70	95.10	94.90	99.70	243
Covid-19	95.70	96.60	96.20	99.50	536
Macro avg.	96.00	95.60	95.77	98.30	4,759

Table 3: Confusion matrix on the held-out test set, from `chest_classifier_metrics.json` for the served checkpoint. The repository `README.md` carries a different matrix that does not match this file on cells other than the headline pneumonia-to-Normal count; the matrix here is the one behind every classification number in this paper.

True (rows) / Pred. (cols)	Normal	Pneumonia	Tuberculosis	Covid-19
Normal	2,465	14	9	14
Pneumonia	102	1,363	4	9
Tuberculosis	10	2	231	0
Covid-19	10	8	0	518

334 The dominant error is one-directional: 102 pneumonia radiographs predicted Normal, 56% of all 182
335 misclassifications, and no other off-diagonal cell exceeds 14. 82 of the 102 come from the COVID-19
336 Radiography Database’s Lung_Opacity folder, which `app.py` maps to Pneumonia though “lung
337 opacity” is broader than pneumonia; 8 come from its Viral Pneumonia folder and 12 from the
338 paediatric pneumonia set. A left–right lung-field mean-intensity probe supports reading this as label
339 noise: median absolute difference 5.9 grey levels on these 102 films, against 8.9 on correctly classified
340 pneumonia and 4.0 on true Normals, and only 22% of the 102 exceed the 90th percentile of the
341 true-Normal distribution, against 37% of correctly classified pneumonia, so the model’s outputs track
342 the pixels rather than miss something present.

343 **Source-bias probe.** An earlier composite drew three of four classes from one source each, so a
344 model could score by recognising the scanner rather than the pathology [Zech et al., 2018]. The

345 composite here weakens that: tuberculosis, Covid-19, and pneumonia each draw from at least two
 346 sources, though the COVID-19 Radiography Database still supplies three classes through one pipeline.
 347 Per-source accuracy spans 83.33% (Montgomery, 12 images) to 99.50% (TBX11K), wider than
 348 prevalence alone explains, though the two extremes are also the two smallest partitions, so sampling
 349 noise and genuine source sensitivity cannot be separated here. The 96.18% headline is read as inflated
 350 by acquisition cues to an unknown degree.

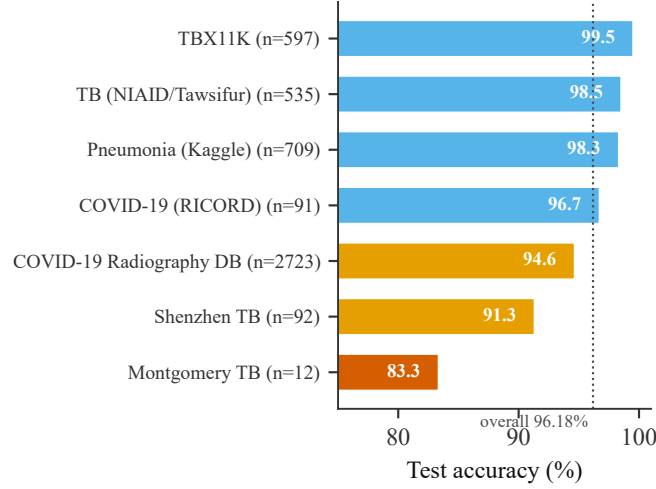


Figure 4: Test accuracy by source dataset, sorted, with support (n) per source and the overall figure marked. The two weakest sources are also the two smallest partitions (92 and 12 images), so this narrows the source-cue confound without settling it.

351 B Report-generation templates, verbatim

352 The complete text-generating logic of the deployed system, app.py lines 1645–1680 (function
 353 predict_image), verbatim. The four descriptions strings and the five heatmap branches are the
 354 entire vocabulary the interface can produce. No line reads a coordinate, area, side, or component
 355 count.

```

356 descriptions = {
357     "Normal": "No abnormal opacities detected in the lung fields.",
358     "Pneumonia": "Findings consistent with pneumonia. Warmer areas indicate possible consolidation.",
359     "Tuberculosis": "Findings consistent with tuberculosis. Warmer areas indicate possible focal lesions",
360     "Covid-19": "Findings consistent with Covid-19. Warmer areas indicate possible bilateral",
361     "": "or cavitation.",
362     "": "ground-glass opacities.",
363 }
364
365
366 if is_uncertain:
367     txt = (f"***Prediction:** Uncertain\n\n"
368           f"The top class is {top_cls} at {top_prob * 100:.1f}%, below the "
369           f"{CONFIDENCE_THRESHOLD * 100:.0f}% reporting threshold. Review the class "
370           f"confidence values; a repeat or higher quality image may help.")
371 else:
372     txt = (f"***Prediction:** {top_cls} ({top_prob * 100:.1f}% confidence)\n\n"
373           f"{descriptions[top_cls]}")
374
375 # Explain what the heatmap represents for this specific case.
376 if mask is None:
377     txt += "\n\nNo attention map is available for this model."
378 elif is_confident_finding:
379     txt += (f"\n\nHeatmap: region driving the {viz_cls} prediction. "
380           f"The outline marks the most influential area.")
381 elif is_auto and top_idx == 0:
382     txt += (f"\n\nHeatmap: areas the model assessed for {viz_cls} "
383           f"(the next most likely class); none reached an abnormal level.")
384 elif is_uncertain:
385     txt += (f"\n\nHeatmap: areas the model weighed for {viz_cls}. "
386           f"Interpret with caution while the prediction is uncertain.")
387 else:

```

```

388     txt += f"\n\nHeatmap: model attention for {viz_cls} ({viz_prob * 100:.1f}%)."
389
390 if mask is not None:
391     txt += ("\n\nOverlay source: disease-segmentation U-Net._"
392           if mask_source == "segmentation"
393           else "\n\nOverlay source: Grad-CAM._")

```

395 CONFIDENCE_THRESHOLD = 0.60 (app.py:67); below it the first branch fires. The only other non-
396 standard path is a caught exception that returns an empty string and a Gradio warning. mask_source
397 is a two-word provenance label naming which network produced the overlay, never what it contains.

398 C Qualitative examples

399 Films from the 245-film pool, S0/S1 text verbatim from tier1_summaries.csv.

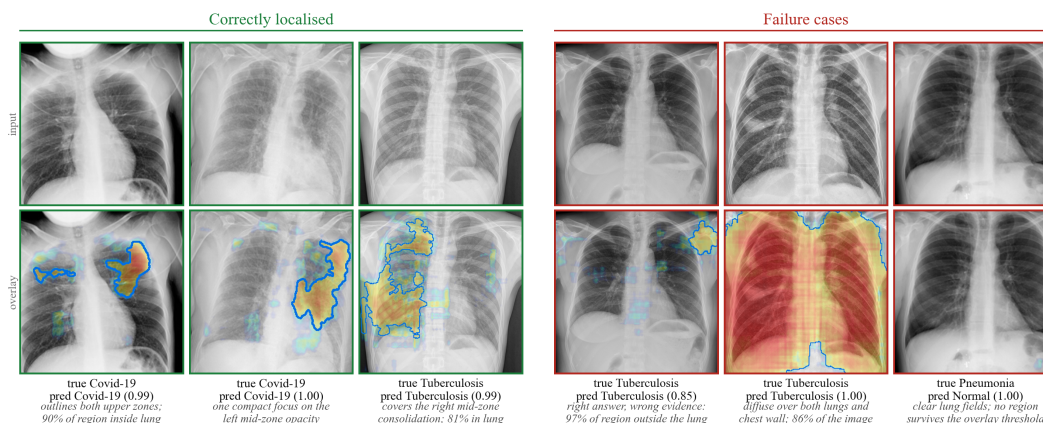


Figure 5: Six served overlays, input above overlay, spanning the range Appendix D quantifies. Left (green): compact regions that settle inside the lung field and follow the visible opacity, 66–90% of the region inside the lung. Right (red): a correct tuberculosis call whose evidence sits almost entirely in soft tissue at the apex (97% outside the lung); a region covering 86% of the image, spanning both lungs and the chest wall; and a clear, well-aerated film the model calls Normal, where no region survives the display threshold at all. All six overlays came from the served U-Net path. Per-case labels give the true class, the prediction, its confidence, and what the region did; italics are this appendix's reading, not part of the served text.

400 **Phantom-outline claims** (both films where is_confident_finding is true but no region survives
401 the threshold; both Montgomery TB group).

- 402 • 194_MCUCXR_0015_0: true Normal, predicted Tuberculosis at 95.7%. S0: "... The outline
403 marks the most influential area. Overlay source: disease-segmentation U-Net." No region
404 exists in the image. S1 under H_actual: "... No heatmap region met the reporting threshold;
405 no location is reported." A confidently wrong call compounded by a claim about an outline
406 that was never drawn.
- 407 • 202_MCUCXR_0301_1: true and predicted Tuberculosis at 94.0%, classification correct,
408 same phantom-outline pattern.

409 **Well-localised.** 209_Normal-2399: true and predicted Normal at 100.0%, single focal region,
410 upper-left zone, 93.6% inside the lung. S1: "... Heatmap evidence: left lung, upper zone, focal,
411 single, consistent with the normal finding."

412 **Diffuse overlay.** 011_person730_virus_1351: true Pneumonia, predicted Normal at 96.0%. The
413 region is extensive and multifocal, centroid outside both lungs, yet 50.3% of its area is inside the lung
414 field, just over the gate. S1: "outside both, mid zone, extensive, multifocal," a literal description of a
415 poorly localised region on a film the classifier got wrong, and a boundary case worth flagging.

416 **No region.** 003_person1175_virus_1981: true Pneumonia, predicted Normal at 81.1%. No
 417 region survives the threshold. S0 gives no hint (“...none reached an abnormal level.”); S1 states
 418 plainly that no location is reported.

419 D Additional qualitative statistics

420 Same 245-film pool. 41 (16.7%) produce no region above the overlay threshold; geometry percentages
 421 are over the remaining 204.

- 422 • *Activation leaving the lung.* 55% of overlays place the majority of the region outside the lung
 423 fields; 20% place almost all of it there, usually on the cardiac silhouette, the mediastinum,
 424 or below the diaphragm. The centroid is on neither lung in 129 of 204.
- 425 • *Edge activation.* 54% of regions reach the outer 2% of the frame despite the 8% crop; 14%
 426 put more than a quarter of the region in the outer tenth.
- 427 • *Diffuse activation.* 44% of regions cover more than a quarter of the image, 19% more than
 428 half, and 37% span both lung fields. This concentrates in Shenzhen (74% of its overlays
 429 diffuse; median TB-positive region 51% of the image); Montgomery fails oppositely (no
 430 region on a quarter of films; median region 4% where one exists).

431 On the 102 pneumonia-to-Normal errors, 28 produce no region and none produces a region over
 432 half the image; median region area is 11%, against 41% for correctly classified pneumonia. This is
 433 consistent with the label-noise reading: the lung fields are frequently clear and the small regions the
 434 model does produce plausibly agree with unremarkable pixels.

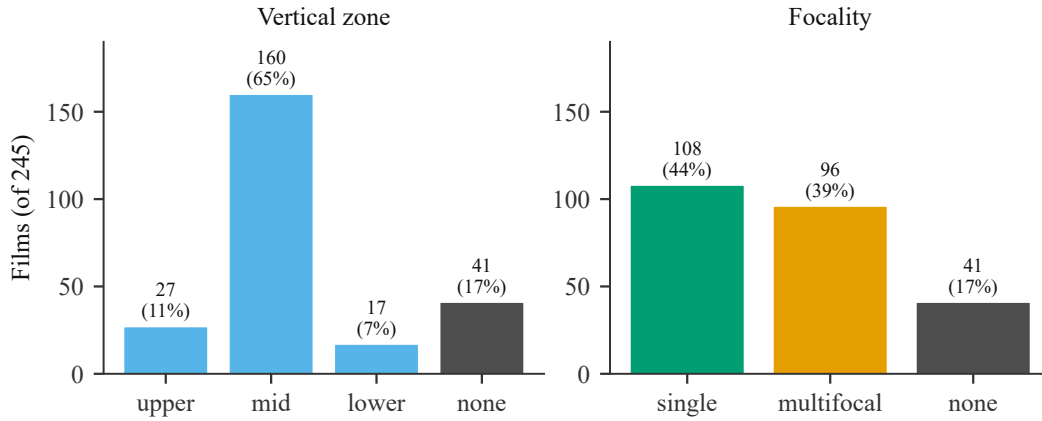


Figure 6: The two region descriptors added for this evaluation (Section 4), over all 245 films. Two thirds of regions centre in the mid zone; most of the rest produce no region at all, rather than an upper- or lower-zone one. Single- and multi-component regions are close to a coin flip, which is why focality is reported as its own descriptor rather than folded into extent.

435 E Latency, per stage

436 Table 4 splits the mean CPU wall-clock time reported in Section 5 by which localisation path served
 437 the request (i7-13620H, 10 PyTorch threads, single request). The two rows are two different code
 438 paths, not one path with variance: the served U-Net returns a mask in one forward pass; the live Grad-
 439 CAM fallback needs a forward pass and a backward pass through the classifier. Report generation
 440 is a dictionary lookup and does not appear as a separate row because it does not register above
 441 measurement noise.

Table 4: CPU latency by localisation path, i7-13620H, 10 threads, single request.

Path	Share of test films	Mean (sd), ms
Distilled U-Net (one forward pass)	91%	293 (81)
Live Grad-CAM fallback (forward + backward)	9%	560 (22)
Weighted (actual test-set mix)	100%	318

NeurIPS paper checklist

1. **Claims.** Do the abstract and introduction state claims that match the paper’s contributions and scope?
[Yes]The central claim, that the deployed summary does not read the mask, is a source-level fact quoted verbatim in Section 3.2 and Appendix B. The abstract and Section 6 state which evaluation-grid cells are measured and which are proposed.
2. **Limitations.** Does the paper discuss the limitations of the work?
[Yes]Section 6: unrun cells (H2, H4, S3/S4), no clinician ratings, dataset-source confounding, label noise in the dominant error mode, and the reproducibility fragility of the split.
3. **Theory.** Are assumptions and proofs given for every theoretical result?
[N/A]The paper reports measurements, not theorems. The one derived bound (localisation quality upper-bounds spatial-claim correctness) is stated with the Dice measurement it rests on.
4. **Experimental reproducibility.** Is there enough information to reproduce the main results?
[Yes]The audited system is public (repository withheld for double blind). The descriptor extractor, the summary reconstructions, the thresholds, and the derangement seed are in Section 4; the 245-film scoring table and mask arrays are the system’s own published analysis outputs.
5. **Open access to data and code.** Are data and code available?
[Yes]The audited system and its analysis outputs are public. No new model was trained.
6. **Experimental settings.** Are training and test details specified?
[Yes]For the components this paper adds there is no training (Section 4). The fixed upstream classifier and U-Net settings are in Section 3 and Appendix A.
7. **Error bars.** Are error bars or other measures of statistical significance reported?
[Yes]Distributions (latency, Dice) carry a mean and standard deviation or a median. Single-number rates are exact counts over the full 245-film pool, so a confidence interval would describe sampling that did not occur.
8. **Compute.** Is the compute for the experiments reported?
[Yes]The analysis is geometry over saved mask arrays and runs in seconds on a CPU; latency figures are from one laptop CPU (i7-13620H, 10 threads). No GPU, no training.
9. **Code of ethics.** Does the research conform to the NeurIPS Code of Ethics?
[Yes]A retrospective audit of an existing research and educational prototype; no patient contact; the interface already carries a non-diagnostic disclaimer this paper argues should be more prominent.
10. **Broader impacts.** Are potential positive and negative societal impacts discussed?
[Yes]Section 6 (Broader impact): the persuasiveness of a label plus a plausible heatmap plus a fluent summary, and why this paper is a caution against it.
11. **Safeguards.** Are safeguards described for any high-risk release?
[N/A]No models, datasets, or scrapers are released.
12. **Licenses.** Are existing assets credited and their licenses respected?
[Yes]Backbone, localisation method, and segmentation architecture are cited in Section 3. The eight source datasets are public Kaggle datasets, named by their exact identifiers so each license can be consulted on its own page:
 - tawsifurrahman/tuberculosis-tb-chest-xray-dataset
 - pcbreviglieri/pneumonia-xray-images

488 • raddar/ricord-covid19-xray-positive-tests
489 • tawsifurrahman/covid19-radiography-database
490 • raddar/tuberculosis-chest-xrays-shenzhen
491 • raddar/tuberculosis-chest-xrays-montgomery
492 • vbookshelf/tbx11k-simplified
493 • legrande/tbdata

494 Terms differ per dataset; the camera-ready should transcribe each from its source page.

495 13. **New assets.** Are new assets documented?
496 [N/A]No new assets are released with this submission.

497 14. **Crowdsourcing and human subjects.** Are crowdsourcing and IRB details given?
498 [N/A]No human subjects, no crowdsourcing, no clinician annotation was collected.

499 15. **IRB / risk for human subjects.** Were risks disclosed and approvals obtained?
500 [N/A]No human-subjects research was conducted.

501 16. **LLM usage.** Is any LLM use in the core method declared?
502 [N/A]No LLM is used in the method. S2/S3/S4 LLM conditions are named as out of scope
503 and not run.